



Neural-Guided Multi-Observer IS-MCTS for Secret Hitler

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INTRODUCTION / PROBLEM

This project investigates whether neural guidance can improve decision-making in Secret Hitler, a challenging hidden-information and deception-heavy game.

We combine multi-observer IS-MCTS with transformer-based state embeddings to capture complex patterns in votes, policies, and player behavior.

The goal is to produce an agent that reasons more realistically about uncertainty than traditional search alone.

KEY FINDINGS

- > Transformer-guided MCTS improves win rate by about 5% at moderate search budgets.
- > The hybrid model achieves up to a 7% win-rate improvement against some opponents.
- > Learned priors provide the largest gains around 100 MCTS iterations.
- > Neural guidance is most beneficial in the three mid- to late-game legislative phases requiring complex inference

BACKGROUND

Secret Hitler features hidden roles, asymmetric information, and strategically motivated misreports, making it a demanding testbed for belief-aware AI.

Players see different parts of the state and must infer intentions from partial observations.

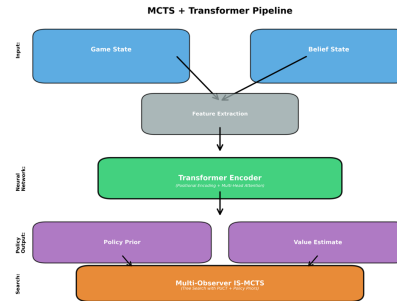
These characteristics require AI systems to model other agents' knowledge and adapt to shifting incentives.

METHODOLOGY

Our agent uses a multi-observer IS-MCTS that assigns each simulated player its own belief distribution, preventing information leakage during search.

A transformer model encodes the evolving game history into a compact representation that guides exploration with learned priors.

The combination enables search to leverage both structured reasoning and data-driven patterns.



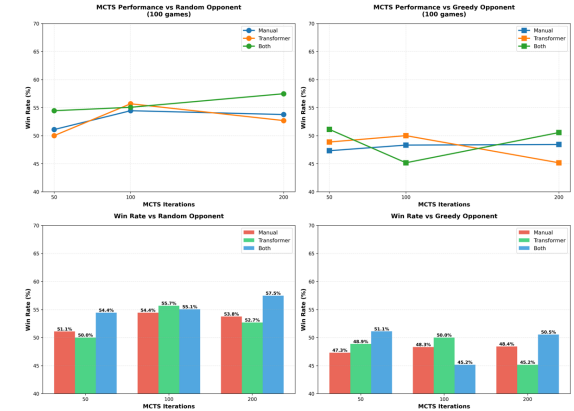
Overall architecture integrating transformer-based state encoding with multi-observer IS-MCTS to guide action selection under hidden information.

RESULTS

Transformer-guided MCTS consistently improves win rates over the manual baseline, especially at moderate search depths around 100 iterations.

Hybrid features achieve the strongest performance in several settings, though results vary depending on opponent strength.

Learned priors help most when computation is limited and the tree is shallow.



CONCLUSION

Learned transformer representations provide actionable guidance that enhances search quality in Secret Hitler.

Multi-observer modeling and neural priors together yield more accurate and human-like decisions under uncertainty.

These results show that combining structured search with learned state embeddings is a promising strategy for hidden-information games.

FUTURE WORK

Future work includes training larger transformer models, improving belief encoders, and exploring lighter statistical methods such as logistic regression for stable priors.

Further tuning of the interaction between neural guidance and MCTS may reduce sensitivity to iteration count.

Extending this framework to other hidden-information games could reveal broader generalization patterns.